

Chapter 1 Curves

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1-2 Parametrized Curve

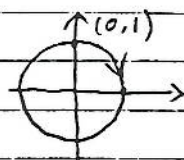
Page 5

#1 Let $\alpha(t) = (\sin t, \cos t)$, then

$$x^2(t) + y^2(t) = \sin^2 t + \cos^2 t = 1$$

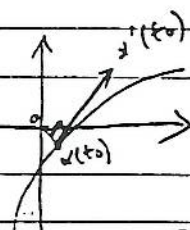
$$\alpha(0) = (0, 1)$$

Q.E.D



#2 Let $l(t) = \|\alpha(t)\|^2$

To minimize $l(t)$ is equivalent to minimize $\|\alpha(t)\|$.



Since $l(t)$ achieves its minimum at t_0 , then $l'(t_0)$ must equal to zero.

$$\text{Now } l(t) = \alpha(t) \cdot \alpha(t)$$

$$l'(t) = \alpha'(t) \cdot \alpha(t) + \alpha(t) \cdot \alpha'(t) = 2\alpha'(t) \cdot \alpha(t)$$

$$\Rightarrow l'(t_0) = 2\alpha'(t_0) \cdot \alpha(t_0) = 0$$

$$\Rightarrow \alpha'(t_0) \cdot \alpha(t_0) = 0$$

$$\Rightarrow \alpha'(t_0) \perp \alpha(t_0)$$

Q.E.D

#3 Suppose $t \in (a, b)$, then $\alpha(t)$ is a line segment where $t \in (a, b)$.

Pf: $\alpha''(t) = 0$

$$\Rightarrow \alpha'(t) = \underline{a} \text{ (constant)}$$

$$\Rightarrow \alpha(t) = \underline{a}t + \underline{b} \text{ (constant)}$$

Q.E.D

#4 Let $\alpha(t) = (x(t), y(t), z(t))$, $t \in I$, $0 \in I$

$$\alpha'(t) = (x'(t), y'(t), z'(t)), \quad V = (a, b, c)$$

$$\text{Now } \alpha'(t) \cdot V = 0$$

$$\Rightarrow a x'(t) + b y'(t) + c z'(t) = 0$$

$$\Rightarrow a \int_0^t x'(t) dt + b \int_0^t y'(t) dt + c \int_0^t z'(t) dt = 0$$

$$\Rightarrow a x(t) \Big|_0^t + b y(t) \Big|_0^t + c z(t) \Big|_0^t = 0$$

$$\Rightarrow a x(t) + b y(t) + c z(t) - a x(0) - b y(0) - c z(0) = 0 \Rightarrow \alpha(t) \cdot V - \alpha(0) \cdot V = 0$$

$\alpha(t) \cdot V$



$$\alpha'(t) \cdot V = 0$$



0

#5 Suppose $|\alpha(t)|$ is nonzero constant.

$$\Rightarrow |\alpha(t)|^2 = c \text{ (constant)}$$

$$\Rightarrow \alpha(t) \cdot \alpha(t) = c$$

$$\Rightarrow \alpha'(t) \cdot \alpha(t) + \alpha(t) \cdot \alpha'(t) = 0$$

$$\Rightarrow 2\alpha'(t) \cdot \alpha(t) = 0$$

$$\Rightarrow \alpha'(t) \perp \alpha(t)$$

Conversely, suppose $\alpha(t) \perp \alpha'(t)$. Let $\alpha(t) = (x(t), y(t), z(t))$

$$\text{Then } x(t)x'(t) + y(t)y'(t) + z(t)z'(t) = 0$$

$$\Rightarrow \int_{t_0}^t x(t)x'(t)dt + \int_{t_0}^t y(t)y'(t)dt + \int_{t_0}^t z(t)z'(t)dt = 0$$

$$\Rightarrow \int_{t_0}^t x(t)dx(t) + \int_{t_0}^t y(t)dy(t) + \int_{t_0}^t z(t)dz(t) = 0$$

$$\Rightarrow \frac{x^2(t)}{2} + \frac{y^2(t)}{2} + \frac{z^2(t)}{2} = \frac{x^2(t_0)}{2} + \frac{y^2(t_0)}{2} + \frac{z^2(t_0)}{2}$$

$$\text{Let } A = x^2(t_0) + y^2(t_0) + z^2(t_0) \text{ (a constant)}$$

$$\Rightarrow x^2(t) + y^2(t) + z^2(t) = A$$

$$\Rightarrow |\alpha(t)|^2 = A \Rightarrow |\alpha(t)| = \sqrt{A}$$

Q.E.D